



KRN-HW-001

Kernellum FPGA Bring-up Protocol

A measurement-first protocol for crossing from ECP5 family synthesis to named-board physical evidence.

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This protocol defines the evidence required before Kernellum may claim physical FPGA implementation, achieved timing, measured inference latency, power, or energy per inference.

The current compiler has an ECP5-85F family target profile and a board-explicit nextpnr scaffold. No board is treated as validated until the device, package, constraints, bitstream, and measurements are recorded.

AI-native systems for efficient computing.



01 / CURRENT EVIDENCE

Current starting point

ITEM	STATUS	NOTE
ECP5 family profile	READY	Lattice ECP5-85F target profile
Yosys synth_ecp5	PASS	Family-mapped synthesis only
Named board	OPEN	Must match hardware actually accessible
LPF constraints	OPEN	Must use real pins and oscillator
nextpnr place-and-route	PENDING	Requires board/package/clock
Measured latency/power	PENDING	Requires physical board

02 / SELECT HARDWARE

Board selection record

FIELD	VALUE TO RECORD	ACCEPTANCE RULE
Board model	_____	Exact commercial/dev board name
FPGA device	_____	Exact ECP5 device
Package	_____	Must match nextpnr package
Oscillator	_____ MHz	Document source / board schematic
Programming path	_____	Tool/cable/interface
Supply method	_____	Needed for power method

03 / IMPLEMENTATION

Constraint and P&R; procedure

```
PACKAGE= LPF= FREQ_MHZ= bash scripts/run_pnr_ecp5.sh
```

- 01 Commit the LPF file used for the run; do not use undocumented pin assignments.
- 02 Capture the exact nextpnr version and complete nextpnr log.
- 03 Record target frequency separately from achieved timing.
- 04 Preserve utilization and timing summaries in a versioned artifacts directory.
- 05 If timing fails, report the failure before changing the target.

04 / TIMING

Timing evidence



MEASUREMENT	RECORDED VALUE	SOURCE
Target clock	_____ MHz	Command / constraints
Worst slack	_____ ns	nextpnr timing report
Achieved / max frequency	_____ MHz	nextpnr timing report
Timing closure	PASS / FAIL	At declared target

Do not substitute the compiler's assumed 100 MHz modeling clock for achieved hardware timing.

05 / BITSTREAM

Board execution

- 01 Generate the bitstream with a tool compatible with the selected board and ECP5 device.
- 02 Program the board and document the command/tool version.
- 03 Use the committed demo top or an equally documented host/UART wrapper.
- 04 Confirm at least one known input produces the expected class/output behavior.
- 05 Repeat after a cold reset to rule out one-off state.

06 / LATENCY

Latency measurement protocol

Preferred measurement: expose a start marker and done marker to GPIO or a logic-analyzer-visible signal. Measure from the accepted start edge to the completion marker. Report clock frequency, sample count, min/median/max latency, and whether input loading/output readback are included.

FIELD	VALUE
Measurement instrument / method	_____
Clock during measurement	_____ MHz
Number of repeated inferences	_____
Core-only latency	_____ μ s
End-to-end latency	_____ μ s
Definition of endpoints	_____

07 / POWER

Power and energy protocol

Measure supply power with a documented instrument. Record idle power and active/inference power under the same board configuration. If a trace is available, integrate power over the inference window; otherwise clearly state the approximation used.



FIELD	VALUE
Instrument	_____
Supply voltage	_____ V
Idle power	_____ W
Active power	_____ W
Energy / inference	_____ J or mJ
Method / integration window	_____

08 / PUBLISH

Acceptance threshold

- 01 Named board/device/package documented.
- 02 Constraint file committed.
- 03 Place-and-route log preserved.
- 04 Timing closure status reported without hiding failure.
- 05 Bitstream demonstrated on the actual board.
- 06 Latency measurement method and endpoint definition documented.
- 07 Power/energy measurement method documented.
- 08 BUILD_STATUS.md, evidence page, hardware page, and next technical report updated from the same measurement record.